

Statements and Truth

Name: _____ Date: _____

Patience rule: Every item below needs a written justification. A label alone (“statement, true”) is not enough — explain *why* it is a statement, or why it is open / not a statement.

Deduction problems — Statements

D1.1 DEDUCTION

Classify each sentence below as exactly one of: **Statement (T)**, **Statement (F)**, **Open sentence**, or **Not a statement**. For every item, write 2–4 sentences of justification. If you need a precise definition to decide (e.g. what counts as a “planet”), state the definition you are using *before* giving T/F.

1. “The integer k satisfies $k^2 < 10$.”
2. “Every even integer greater than 2 is the sum of two primes.”
3. “Stop cheating on the quiz.”
4. “This sentence contains exactly eight words.”
5. “The temperature in Melbourne at noon on 1 January 2025 was above 30°C .”
6. “Either team A wins or team B wins, but not both.”
7. “Why does ice float?”
8. “For all real numbers x , if $x > 3$ then $x^2 > 9$.”
9. “The movie was boring.”
10. “There exists an integer n such that $n^2 = 2$.”

TEN CLASSIFICATIONS WITH JUSTIFICATIONS (USE NUMBERED SUB-ANSWERS)

D1.2 DEDUCTION

Fix the domain to **integers from 1 through 12**. Let $P(n)$ = “ n is prime,” $Q(n)$ = “ n has exactly two positive divisors,” $R(n)$ = “ n is a perfect square.”

For each n in $\{1, 2, 4, 6, 9, 11\}$, do all of the following:

- State whether $P(n)$, $Q(n)$, and $R(n)$ are open or closed once n is fixed.
- Give the truth value of each closed sentence.
- Decide whether “ $P(n)$ if and only if $Q(n)$ ” is true for that fixed n , with one sentence of reason.

You must process all six values of n — no skipping “obvious” ones.

SIX BLOCKS (ONE PER n) WITH (A)(B)(C)

D1.3

DEDUCTION

A teacher hides an integer x with $1 \leq x \leq 10$. Four students make claims:

- **Ada:** “ x is even and $x > 5$.”
- **Ben:** “ x is odd or $x = 4$.”
- **Cara:** “It is not true that $x \leq 3$.”
- **Dan:** “ x is a prime number.”

You are told: **exactly three** of the four claims are true statements for the hidden x , and **exactly one** is false.

1. Rewrite each student's claim as a closed statement by replacing x with a variable over $\{1, \dots, 10\}$ (already done above — now treat them as formulas).
2. For each candidate x from 1 to 10, record T/F for all four claims in a table (10 rows).
3. Circle every x that satisfies the “exactly three true” rule.
4. Prove whether the solution for x is unique. If not unique, list all possibilities and explain what extra information would decide between them.

TABLE + CONCLUSION + UNIQUENESS ARGUMENT

D1.4 DEDUCTION

Consider the sentence: “The 100th decimal digit of π is 7.”

1. Is this a statement in classical logic? Argue both sides briefly, then state which side you accept and why.
2. Now consider: “The 100th decimal digit of π will be 7 when someone computes it.” Does changing the tense change your classification? Explain in 4–6 sentences.
3. Write your own example of a sentence that *sounds* like a statement about the future but is *not* a statement under the “fixed right now” standard. Then write a closely related sentence that *is* a statement because the standard is pinned down.

THREE-PART WRITTEN ANALYSIS